WORKED SOLUTION

STEP III 2008, Question 1

Pure • Non-linear Simultaneous Equations

> **STEP III 2008, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/08-s3-q1

The key idea

Write $s_k = ax^k + by^k$, so the equations give s_0, s_1, s_2, s_3. Since x and y are the roots of $t^2 - pt + q = 0$, where $p = x + y$ and $q = xy$, the power sums obey a recurrence: each s_k is p times the previous one minus q times the one before that. That is exactly what the hint's "multiply by $x + y$" produces. The four non-linear equations become two *linear* equations in p and q. After that, a quadratic gives x and y, and the first two equations give a and b.

Solution

Let $s_k = ax^k + by^k$ for $k = 0, 1, 2, 3$, so that $s_0 = 1$, $s_1 = \frac{1}{3}$, $s_2 = \frac{1}{5}$, $s_3 = \frac{1}{7}$. Put $p = x + y$ and $q = xy$.

A recurrence for the power sums

For any a, b, x, y and any $k \geq 1$,

$$(x + y)(ax^k + by^k) = ax^{k+1} + by^{k+1} + xy\,(ax^{k-1} + by^{k-1}),$$

because the cross terms are $axyx^{k-1}$ and $bxy\,y^{k-1}$. In our notation,

$$s_{k+1} = p\,s_k - q\,s_{k-1}.$$

This is an identity, so it holds for *every* solution of the system.

Finding $x + y$ and xy

Taking $k = 1$ and $k = 2$ and substituting the known values:

$$\tfrac{1}{5} = \tfrac{1}{3}p - q,$$
$$\tfrac{1}{7} = \tfrac{1}{5}p - \tfrac{1}{3}q.$$

From the first, $q = \frac{1}{3}p - \frac{1}{5}$. Substituting into the second,

$$\tfrac{1}{7} = \tfrac{1}{5}p - \tfrac{1}{9}p + \tfrac{1}{15} \quad \Longrightarrow \quad \tfrac{4}{45}p = \tfrac{1}{7} - \tfrac{1}{15} = \tfrac{8}{105} \quad \Longrightarrow \quad p = \tfrac{6}{7},$$

and then $q = \frac{2}{7} - \frac{1}{5} = \frac{3}{35}$. The linear system has non-zero determinant, so these values are forced: *every* solution has $x + y = \frac{6}{7}$ and $xy = \frac{3}{35}$.

Finding x and y

So x and y are the two roots of

$$t^2 - \tfrac{6}{7}t + \tfrac{3}{35} = 0, \qquad \text{that is,} \qquad 35t^2 - 30t + 3 = 0,$$

which gives

$$t = \frac{30 \pm \sqrt{900 - 420}}{70} = \frac{30 \pm 4\sqrt{30}}{70} = \frac{15 \pm 2\sqrt{30}}{35}.$$

The discriminant is positive, so $x \neq y$, and $\{x, y\}$ is this pair in one order or the other.

Ask yourself Can we be sure a and b are determined once we know x and y?

Yes. With x and y known, the first two equations, $a + b = 1$ and $ax + by = \frac{1}{3}$, are linear in a and b with determinant $y - x$. That is non-zero because $x \neq y$, so there is exactly one pair (a, b).

Finding a and b

Take $x = \dfrac{15 + 2\sqrt{30}}{35}$ and $y = \dfrac{15 - 2\sqrt{30}}{35}$, so $x - y = \dfrac{4\sqrt{30}}{35}$. Eliminating b from $a + b = 1$ and $ax + by = \frac{1}{3}$ gives $a(x - y) = \frac{1}{3} - y$. Now

$$\tfrac{1}{3} - y = \frac{35 - 3(15 - 2\sqrt{30})}{105} = \frac{6\sqrt{30} - 10}{105},$$

so

$$a = \frac{6\sqrt{30} - 10}{105} \cdot \frac{35}{4\sqrt{30}} = \frac{6\sqrt{30} - 10}{12\sqrt{30}} = \frac{1}{2} - \frac{5}{6\sqrt{30}} = \frac{1}{2} - \frac{\sqrt{30}}{36},$$

and $b = 1 - a = \dfrac{1}{2} + \dfrac{\sqrt{30}}{36}$.

These really are solutions

We have only shown that any solution must look like this, so we still have to check that it works. By construction $s_0 = 1$ and $s_1 = \frac{1}{3}$. Since p and q satisfy the two linear equations above, the recurrence gives $s_2 = ps_1 - qs_0 = \frac{1}{5}$ and then $s_3 = ps_2 - qs_1 = \frac{1}{7}$. So all four equations hold.

Swapping the roles of x and y swaps a and b. So there are exactly two solutions:

$$(a, b, x, y) = \left(\frac{18 - \sqrt{30}}{36}, \frac{18 + \sqrt{30}}{36}, \frac{15 + 2\sqrt{30}}{35}, \frac{15 - 2\sqrt{30}}{35} \right)$$

and the same with $a \leftrightarrow b$ and $x \leftrightarrow y$ swapped together. In both, the larger of x, y is paired with the smaller coefficient.

Checks

Numerically, $x \approx 0.741556$, $y \approx 0.115587$, $a \approx 0.347855$ and $b \approx 0.652145$. Then $ax + by \approx 0.333333$, $ax^2 + by^2 \approx 0.200000$ and $ax^3 + by^3 \approx 0.142857$, which agree with $\frac{1}{3}$, $\frac{1}{5}$ and $\frac{1}{7}$ to machine precision. A quick sanity check: x and y both lie in $(0, 1)$, and a and b are both positive. That fits the right-hand sides being averages of t^{2k} over $[0, 1]$ (see below).

Common slips

- Solving for x and y but forgetting that the pairing matters: a goes with x, so you can't swap x and y without also swapping a and b.
- Stopping after finding necessary conditions. The question says "find all", so you need both that every solution has this form and that these values do satisfy all four equations.

Beyond STEP

The numbers $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}$ are $\frac{1}{2}\int_{-1}^{1} t^{2k}\,\mathrm{d}t$. So the system asks for a rule $\frac{1}{2}\int_{-1}^{1} f \approx \frac{a}{2}\big(f(\sqrt{x}) + f(-\sqrt{x})\big) + \frac{b}{2}\big(f(\sqrt{y}) + f(-\sqrt{y})\big)$ that is exact for polynomials up to degree 7. This is four-point *Gauss–Legendre quadrature*: its nodes are $\pm\sqrt{x}$ and $\pm\sqrt{y}$, and $a \approx 0.3479$ and $b \approx 0.6521$ are its famous weights. Finding a recurrence from power sums, as we did here, is also the idea behind Prony's method in signal processing. For more, try Süli and Mayers, *An Introduction to Numerical Analysis*.

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